

Statistical Programming Languages

Exercises: Day 3

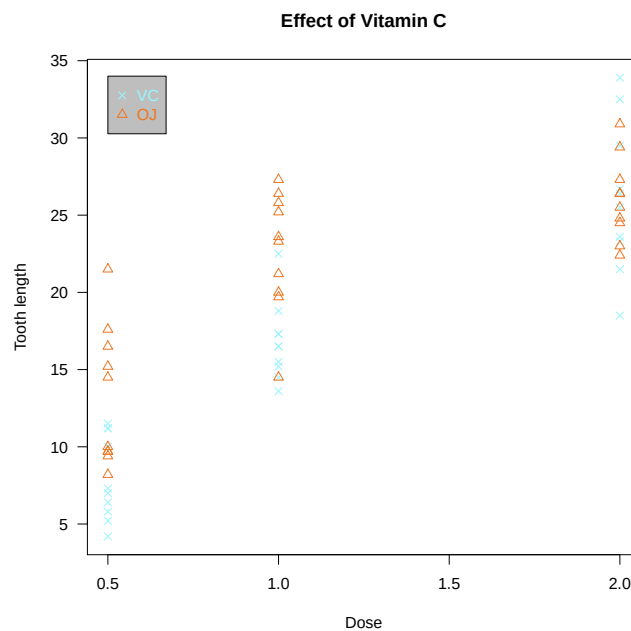
Graphics

Exercise 1 (Testing plot parameters):

On the `pressure` data set try the different plot parameters mentioned on slides 6 to 8 of “6_graphics.pdf”.

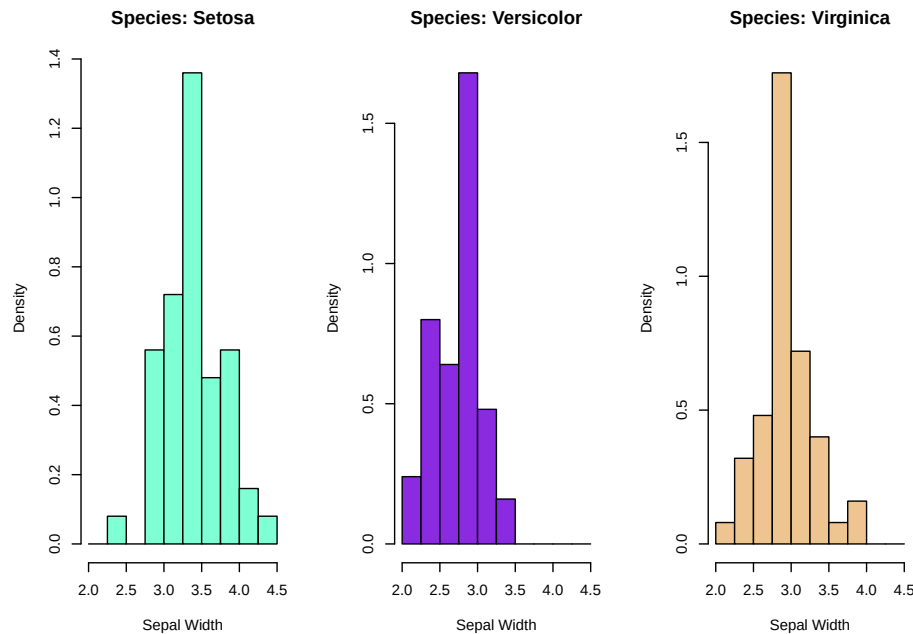
Exercise 2 (Distinguish variables):

Construct the following figure based on the built-in data set `ToothGrowth`, see `?ToothGrowth`. Use the colors `cadetblue1` and `chocolate2`.



Exercise 3 (Side by side plots):

Construct the following figure based on the built-in data set `iris`. Use the colors `aquamarine`, `blueviolet` and `burlywood2`.

**Exercise 4 (ggplot2):**

The built-in data set `cars` gives the speed in mph (`speed`) and stopping distance in ft (`dist`) for 50 cars.

- Use the package `ggplot2` to construct a scatterplot of `speed` (horizontal) against `dist` (vertical).
- Revise the plot by labeling the horizontal axis with “Speed (mpg)” and the vertical axis with “Stopping Distance (ft)”. Add a meaningful title to the plot.

Exercise 5 (Multivariate plots):

Consider the multivariate function $f : \mathbb{R} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}, f(x, y) := \sin(x) + \sqrt{y}$.

- Create a contour plot for $0 \leq x \leq 5$ and $0 \leq y \leq 5$.
- Create a 3D plot for $0 \leq x \leq 5$ and $0 \leq y \leq 5$.